Level 3 Numeracy, Week 3: Understanding Division

Sharing erasers fairly, trading a hundred for ten tens, and seeing long division with blocks

FOR VOLUNTEERS
Make sure every child can do Activity 1 (Q1 and Q2) before you go on to the place-value division. Division is the hardest of the four operations; it is fine to spend two sessions on it.

1. The words of division

Word	Bengali	In 10 ÷ 3 = 3 remainder 1
Dividend	ভাজ্য	10, the amount to be shared
Divisor	ভাজক	3, the number we divide by
Quotient	ভাগফল	3, how many each gets
Remainder	ভাগশেষ	1, what is left over

KEY IDEA
Dividend = divisor × quotient + remainder. Check: 3 × 3 + 1 = 10. This is how we check every division.

2. Activity 1: sharing erasers

Three friends, Inky, Pinky and Ponky, are fighting over 10 erasers. Help them share fairly. Give one eraser to each, then another to each, and keep going until fewer erasers are left than friends.

Round	Inky	Pinky	Ponky	Left over	Can we go again?
1	1	1	1	10 − 3 = 7	Yes
2	1	1	1	7 − 3 = 4	Yes
3	1	1	1	4 − 3 = 1	No (1 < 3)

Each friend gets 3 erasers and 1 is left. So 10 ÷ 3 = 3, remainder 1. Dividend 10, divisor 3, quotient 3, remainder 1.

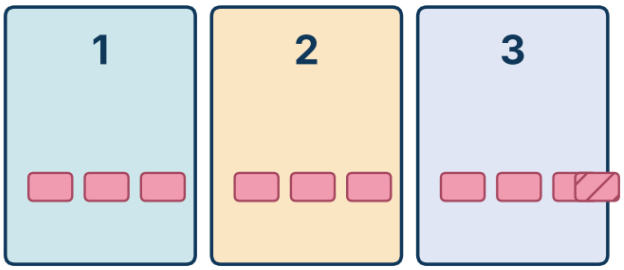


Figure 1. 10 erasers shared among 3 friends: 3 each and 1 left.

Question 2: groups of different sizes

Lay out 10 erasers. Form groups of different sizes and fill in the table. Look at it two ways: sharing (divide among a number of friends) and grouping (make groups of a given size).

Number of groups	Erasers in each group	Dividend	Divisor	Quotient	Remainder
5	2	10	5	2	0
3	3	10	3	3	1
10	1	10	10	1	0
2	5	10	2	5	0
1	6	10	6	1	4

NOTE
Careful with the last row. One group of 6 uses 6 erasers, and 4 are left over: 10 ÷ 6 = 1 remainder 4. The divisor is the group size (6), not the number of groups. We kept the table consistent: the divisor is the number we divide by.

ANSWER KEY (VOLUNTEERS ONLY)
The table above is the answer key. Each row follows dividend = divisor × quotient + remainder: 5 × 2 + 0, 3 × 3 + 1, 10 × 1 + 0, 2 × 5 + 0, 6 × 1 + 4.

3. Activity 2: division with place value

Revise place value with money

Break each number into 100-, 10- and 1-rupee notes and coins:

Number	Thousands	Hundreds	Tens	Ones
121	0	1	2	1

Number	Thousands	Hundreds	Tens	Ones
416	0	4	1	6
4473	4	4	7	3

KEY IDEA
One 100-rupee note = ten 10-rupee notes. One 10-rupee note = ten 1-rupee coins. Moving a 1 one place to the right makes it ten of the smaller thing. This is the whole secret of long division.

Easy first: 48 ÷ 4

Show 4 tens and 8 ones. Share among 4 groups: each group gets 1 ten and 2 ones, so 48 ÷ 4 = 12. Then try 3609 ÷ 3 with blocks: 3 thousands, 6 hundreds, 0 tens, 9 ones. Each of the 3 groups gets 1 thousand, 2 hundreds, 0 tens and 3 ones: 1203.

When it does not fit: 192 ÷ 8

1. Show **1 hundred, 9 tens, 2 ones**. We want groups of 8.
2. There is only 1 hundred, so we cannot make a group of 8 from it. **Trade the hundred for 10 tens.** Now there are **19 tens**.
3. Share 19 tens among 8: **2 tens each** (16 used), **3 tens left over**.
4. Trade the 3 tens for 30 ones, add the 2 ones: **32 ones**. Share 32 among 8: **4 ones each**, nothing left.
5. Each group has 2 tens and 4 ones. **192 ÷ 8 = 24.** Check: 8 × 24 = 192 ✔.

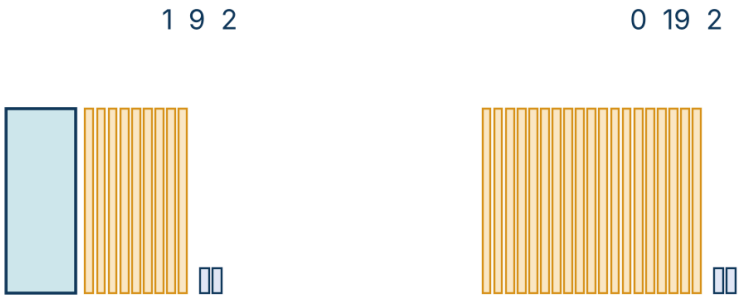


Figure 2. 192 as blocks: 1 hundred, 9 tens and 2 ones. Trade the hundred for 10 tens to make groups of 8.

A bigger divisor: 121 ÷ 11

We must make 11 equal groups from 1 hundred, 2 tens and 1 one. Trade the hundred for 10 tens: 12 tens. Each of the 11 groups takes 1 ten, which leaves 1 ten. Trade it for 10 ones: with the 1 one that is 11 ones. Each group takes 1 one. Every group has 1 ten and 1 one. 121 ÷ 11 = 11.

How the written method matches

Step in 192 ÷ 8	With blocks	On paper
1 hundred is too small; look at 19 tens	trade 1 hundred for 10 tens	19 ÷ 8 = 2
2 tens each uses 16 tens; 3 tens left	write 2 in the tens place	19 − 16 = 3
3 tens and 2 ones make 32 ones	trade 3 tens for 30 ones	bring down the 2: 32
4 ones each uses 32; none left	write 4 in the ones place	32 ÷ 8 = 4, remainder 0

TRY IT

Challenge pairs: 4473 ÷ 213 and 1346 ÷ 112. Use blocks or money, and check by multiplying.

ANSWER KEY (VOLUNTEERS ONLY)

- 48 ÷ 4 = 12; 3609 ÷ 3 = 1203; 192 ÷ 8 = 24; 121 ÷ 11 = 11.
- 4473 ÷ 213 = **21** (213 × 21 = 4473, remainder 0).
- 1346 ÷ 112 = **12, remainder 2** (112 × 12 = 1344, and 1346 − 1344 = 2).

TIP

The reference named in the original is "The Power of the Visual: Seeing Long Division". We could not open it, so we wrote the steps ourselves. Look at it if you can, and share any picture that helps.

Student worksheet

Name: _____	Date: _____

১) একটা গুরুতর সমস্যা! ইঙ্কি, পিঙ্কি আর পঙ্কি ১০টা ইরেজার নিয়ে মারামারি করছে। ন্যায্য ভাগাভাগি করতে একটু সাহায্য করি। ছকটা পূরণ করো:
1) A serious problem! Inky, Pinky and Ponky are fighting over 10 erasers. Let us share fairly. Fill in the table:

দফা	ইঙ্কি	পিঙ্কি	পঙ্কি	বাকি	আবার দেওয়া যাবে?
১					
২					
৩					

ভাজ্য ____, ভাজক ____, ভাগফল ____, ভাগশেষ ____
Dividend ____, divisor ____, quotient ____, remainder ____

২) ১০টি ইরেজার দিয়ে বিভিন্ন সাইজের দল বানাও আর ছকটা ভরো:
2) Make groups of different sizes with 10 erasers and fill in the table:

মোট কতোগুলো দল?	প্রতি দলে কয়টা?	ভাজ্য	ভাজক	ভাগফল	ভাগশেষ
৫	২	১০	৫	২	০
৩					
১০					
২					
	৬	১০			

৩) পুরনো পড়া: সংখ্যাগুলোকে হাজার, শতক, দশক, একক ঘরে ভাঙো:
3) Revision: break each number into thousands, hundreds, tens, ones:

সংখ্যা	হাজার	শতক	দশক	একক
১২১	০	১	২	১
৪১৬				
৪৪৭৩				

৪) ভাগ করা মানে দলে ভাগ করা। ব্লক বা টাকা সাজিয়ে করো:
4) Division means sharing into groups. Use blocks or money:

৪৮ ÷ ৪ = _____	৩৬০৯ ÷ ৩ = _____

৫) ১৯২ ÷ ৮। ১টা শতক থেকে ৮-এর দল হচ্ছে না, তাই শতককে ১০ দশকে ভেঙে ১৯ দশক পাই। প্রত্যেকে ___ দশক পায়, বাকি ___ দশক। বাকি দশক ভেঙে ___ একক পাই, তার সঙ্গে ২ একক = ___ একক। প্রত্যেকে ___ একক পায়। ১৯২ ÷ ৮ = _____
5) 192 ÷ 8. Fill in the blanks as you trade the hundred for tens and the tens for ones.

৬) ১২১টা ইরেজার ১১টা দলে সমান ভাগ করো। প্রতি দলে কয়টা? _____
6) Share 121 erasers equally into 11 groups. How many in each group?

৭) নিজে করো: ৪৪৭৩ ÷ ২১৩ = _____ ১৩৪৬ ÷ ১১২ = _____ (ভাগশেষ ____)
7) You try: 4473 ÷ 213 and 1346 ÷ 112 (remainder?)

৮) গুণ করে মিলিয়ে নাও: ভাজক × ভাগফল + ভাগশেষ = ভাজ্য?
8) Check by multiplying: divisor × quotient + remainder = dividend?